Clynes

Dynamics of Vertex Evoked Potentials: The R-M Brain Function

Original Clynes results

Hypothesis: vertex evoked potentials are elicited when a sensory variable leaves the state of rest and enters the state of motion. This is the **rest-motion (R-M) function.**

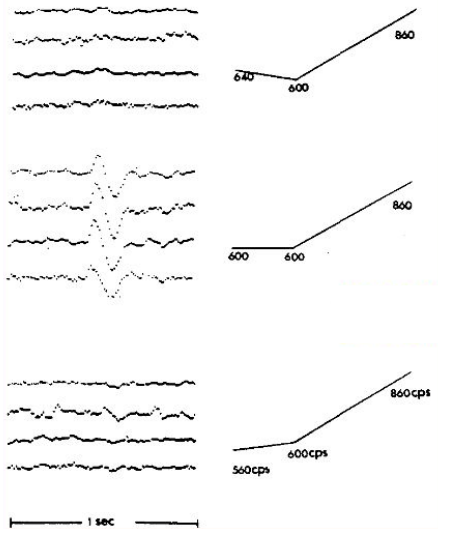


FIGURE F–3.—Inhibition of the R–M response (center) to change in pitch by a previous change in pitch in either direction. The stimulus pattern shown on the right corresponds to the particular group of traces to which it is opposite and is on the same time scale.

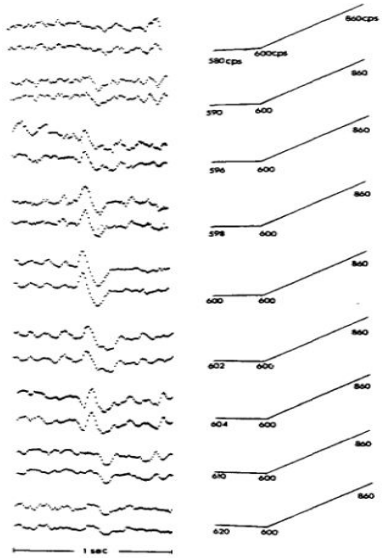
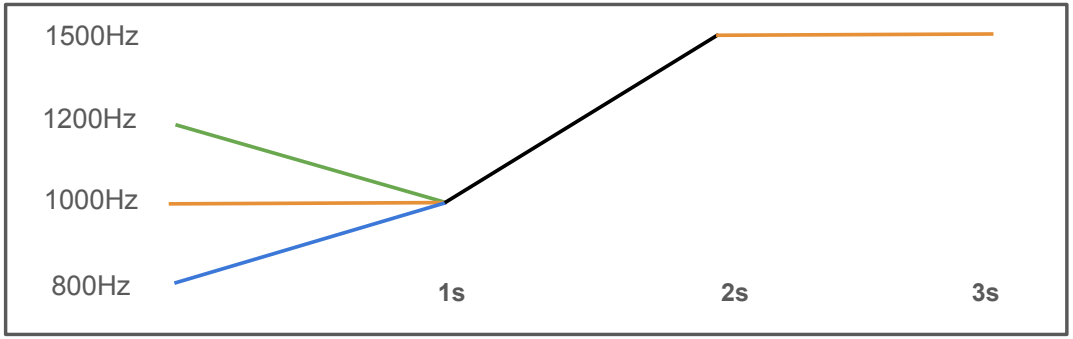


FIGURE F–4.—Gradually increasing inhibition of the R–M response to pitch modulation by previous changes of pitch in either direction. Note the very sharp inhibition produced by small previous changes in pitch.

Experiment 1: Replication of Clynes

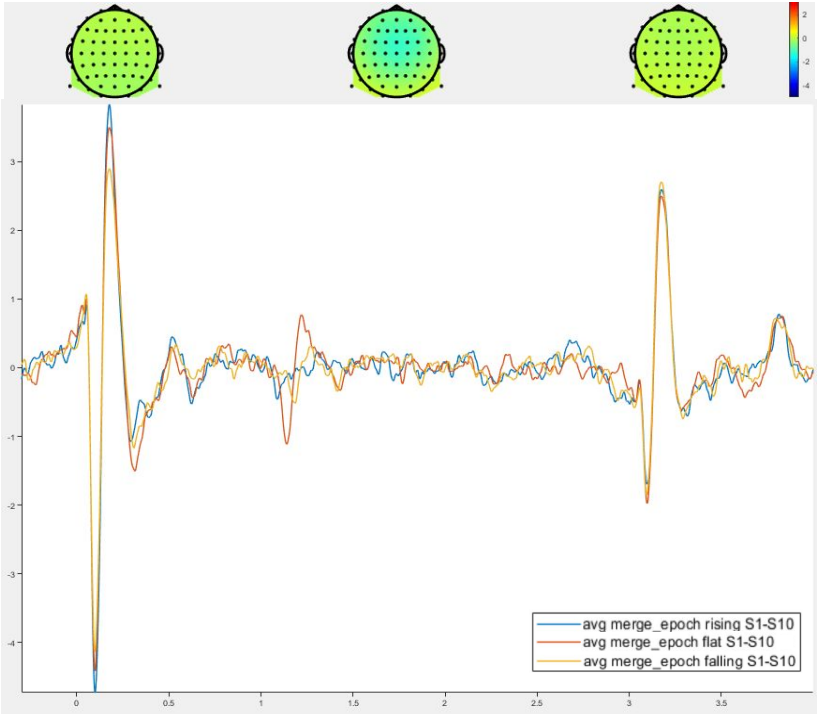


	T0-1 (**0s**)	T0 (**1s**)	T1 (**2s**)
Flat	1000 Hz	1000 Hz	1500 Hz
Rising	800 Hz	1000 Hz	1500 Hz
Falling	1200 Hz	1000 Hz	1500 Hz

19 subjects

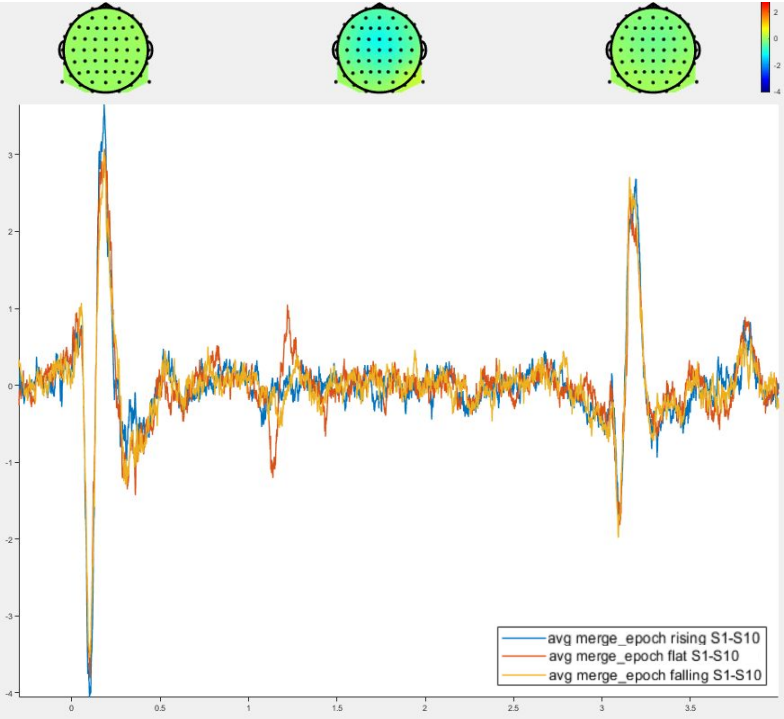
Mean

RISING FLAT FALLING

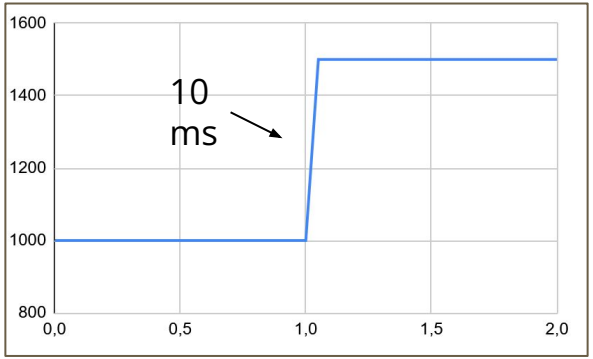


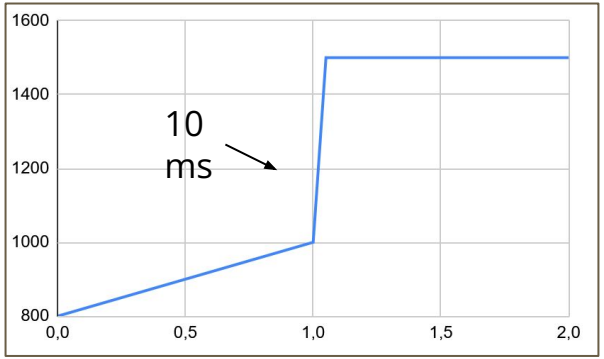
Median

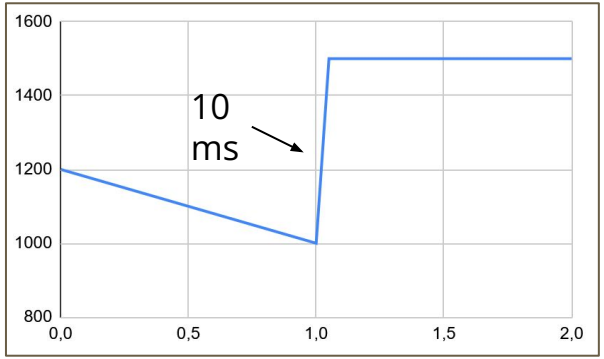
RISING FLAT FALLING



Experiment 2: Fast rise modulation

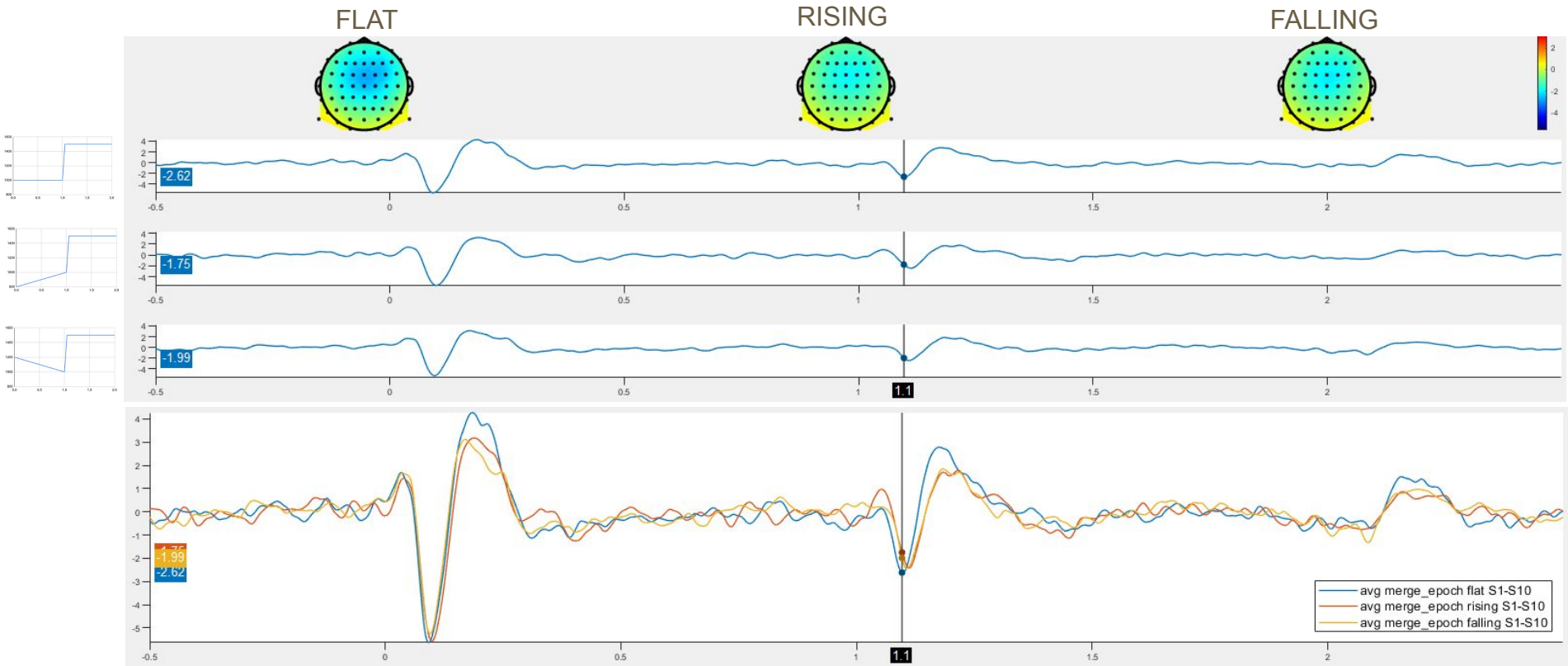




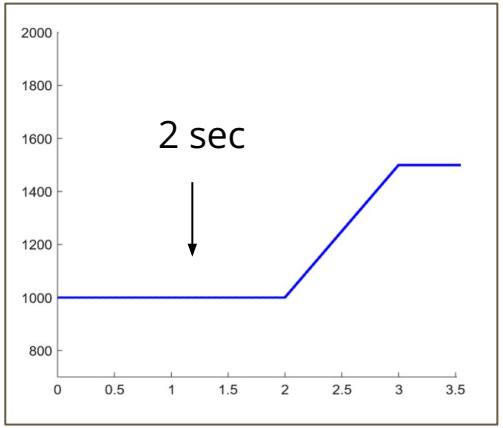


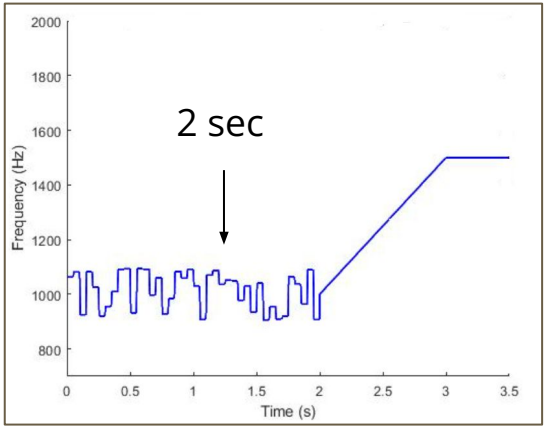
	T0-1 (**0s**)	T0 (**1s**)	T1 (**1.01s**)
Flat	1000 Hz	1000 Hz	1500 Hz
Rising	800 Hz	1000 Hz	1500 Hz
Falling	1200 Hz	1000 Hz	1500 Hz

Mean of ten subjects



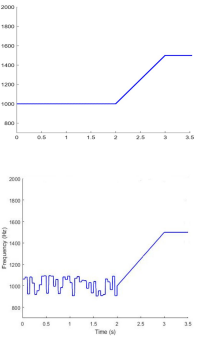
Experiment 3: Scrambled modulation

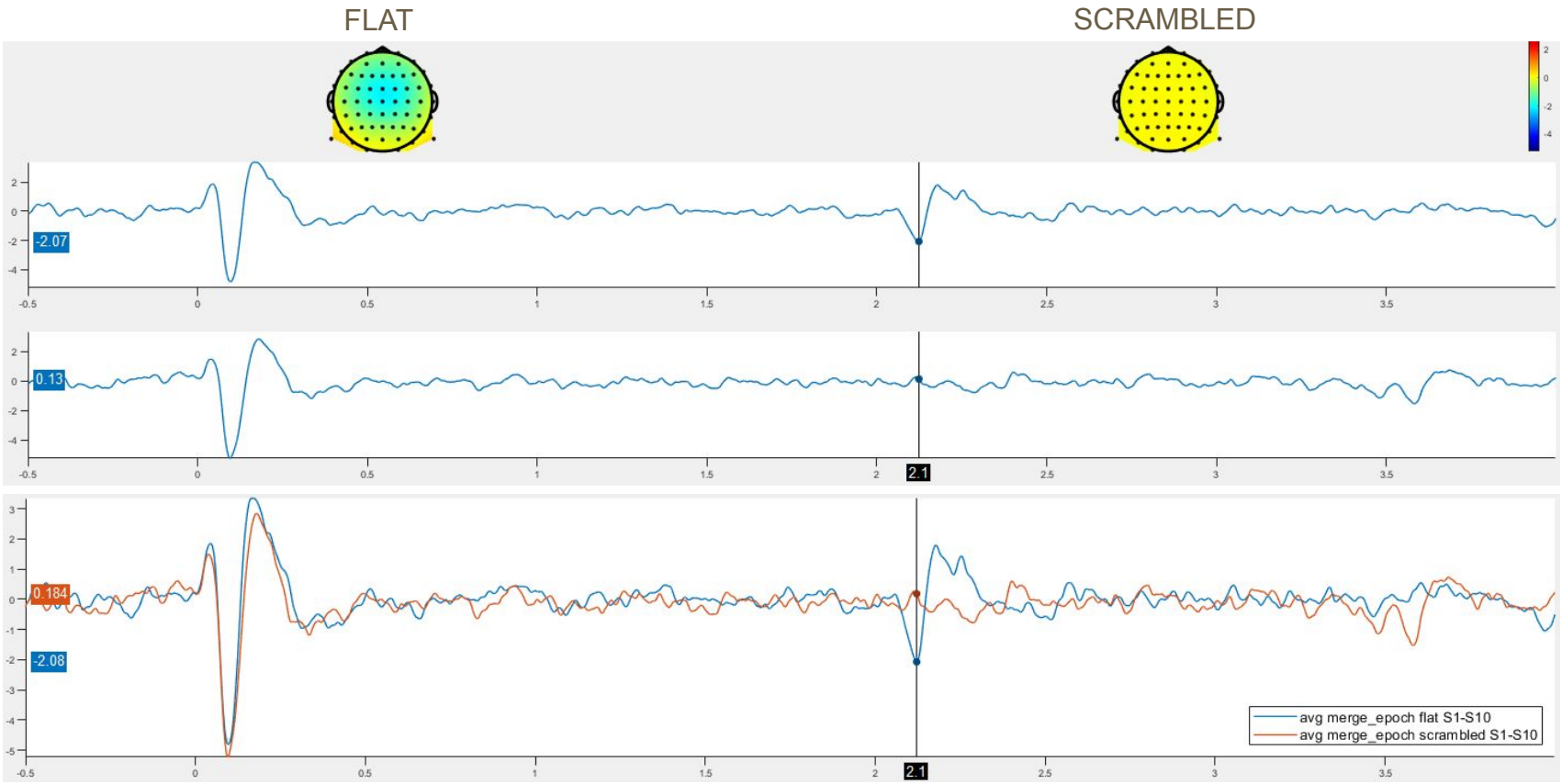




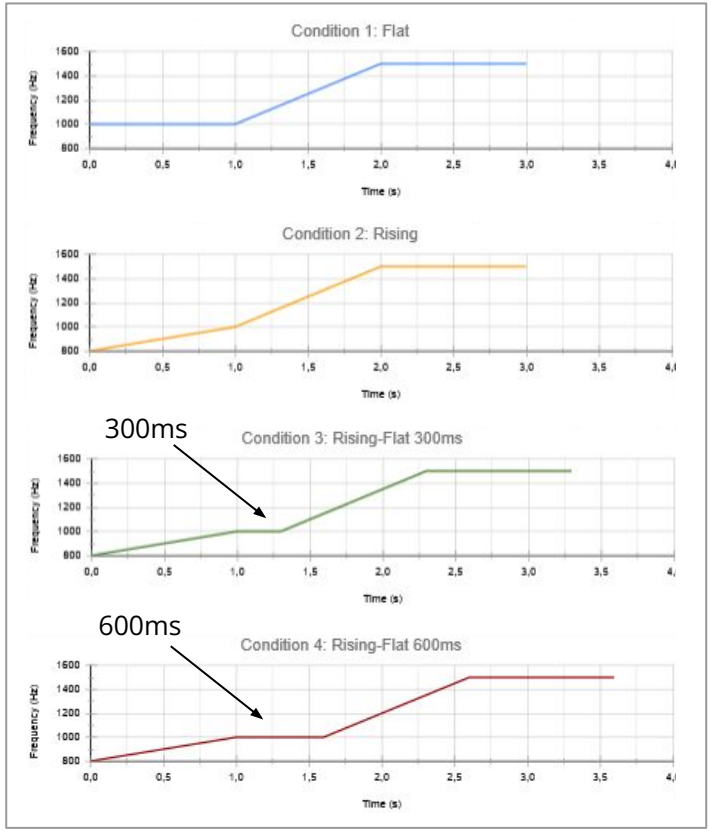
	T0-1 (**0s**)	T0 (**2s**)	T1 (**3s**)
Flat	1000 Hz	1000 Hz	1500 Hz
Scrambled	900-1100 Hz	1000 Hz	1500 Hz

Mean of ten subjects



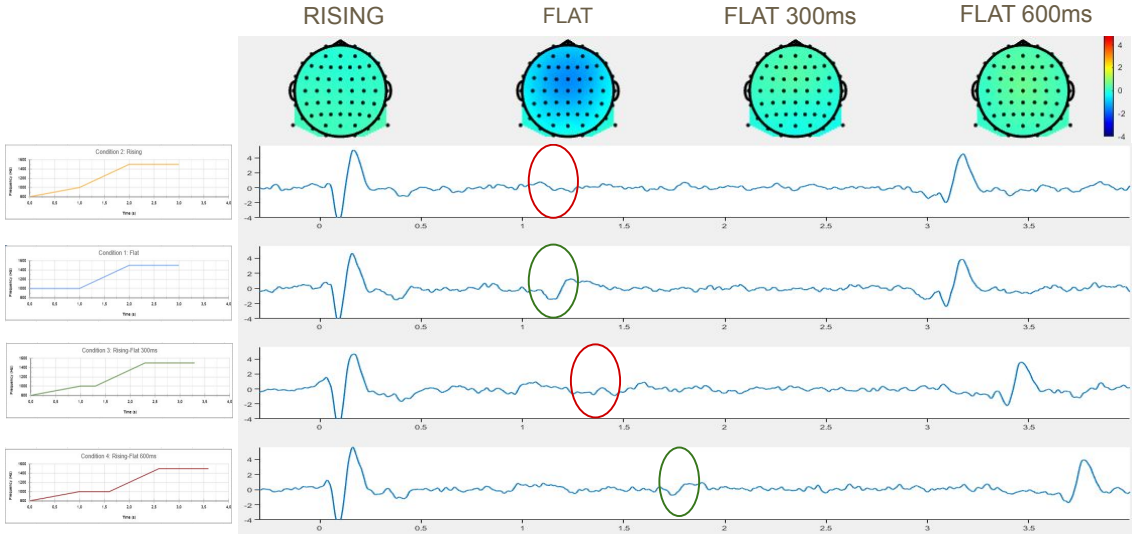


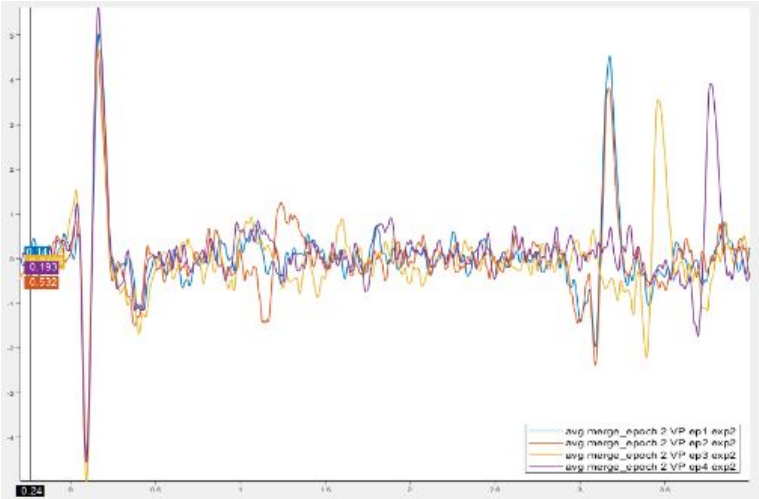
Experiment 4: Modulation of flat



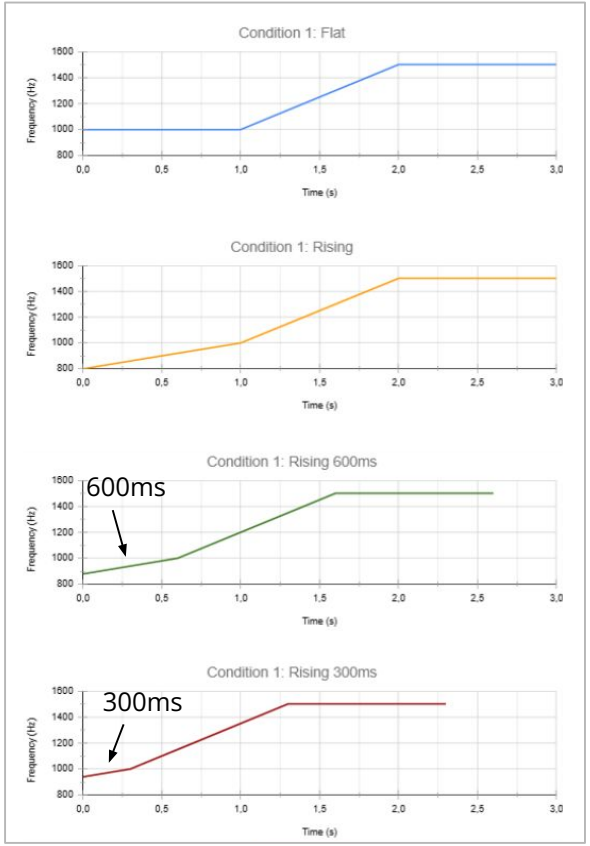
	T-1 (**0s**)	T0	T1
Flat	1000 Hz	1000 Hz (**1s**)	1500 Hz (**2s**)
Rising	800 Hz	1000 Hz (**1s**)	1500 Hz (**2s**)
300ms Flat	800 Hz	1000 Hz (**1.3s**)	1500 Hz (**2.3s**)
600ms Flat	800 Hz	1000 Hz (**1.6s**)	1500 Hz (**2.6s**)

Mean of ten subjects



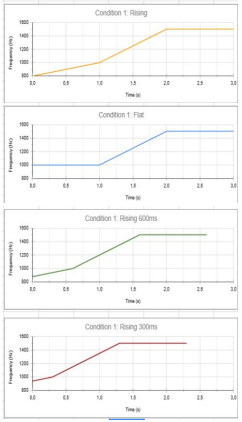


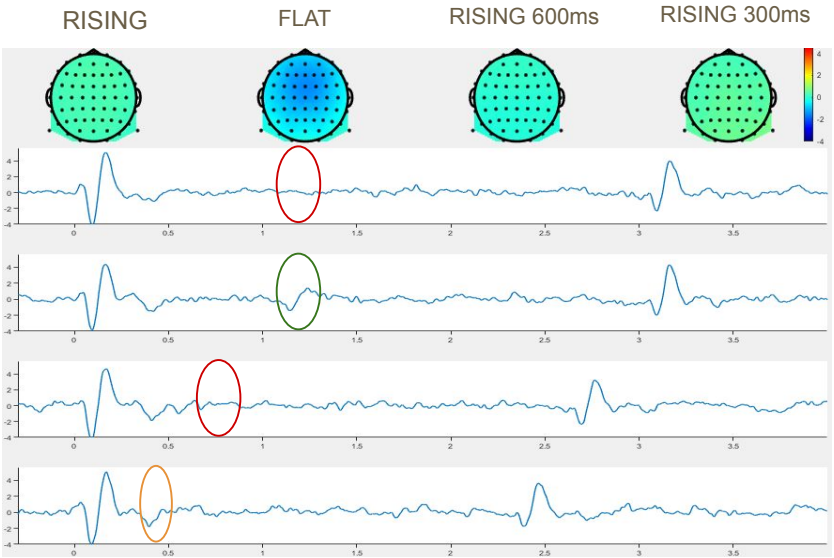
Experiment 5: Modulation of rising

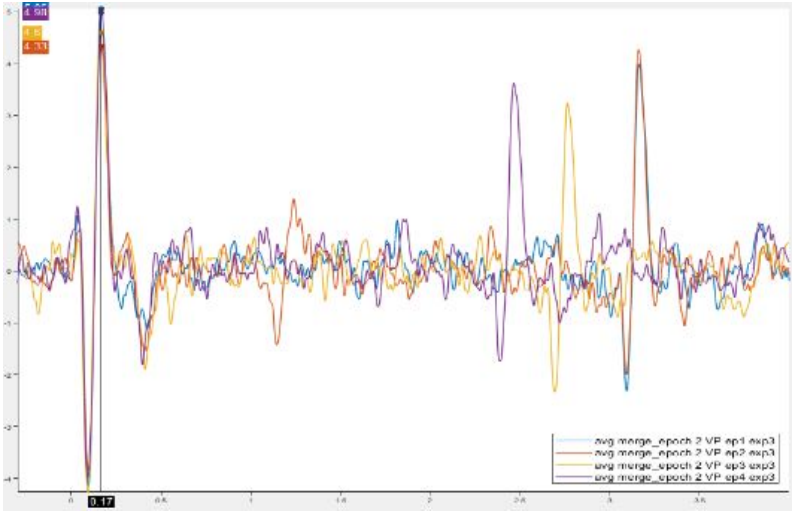


	T0-1 (0s)	**T0**	**T1**
Flat	1000Hz	1000Hz (**1s**)	1500Hz (**2s**)
Rising	800Hz	1000Hz (**1s**)	1500Hz (**2s**)
600ms Rising	880Hz	1000Hz (**0.6s**)	1500Hz (**1.6s**)
300ms Rising	940Hz	1000Hz (**0.3s**)	1500Hz (**1.3s**)

Mean of ten subjects

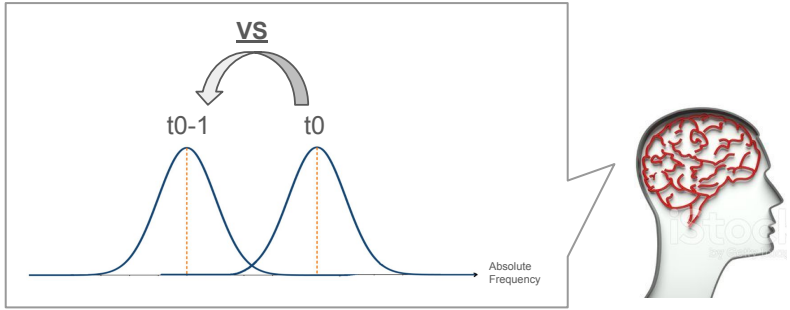




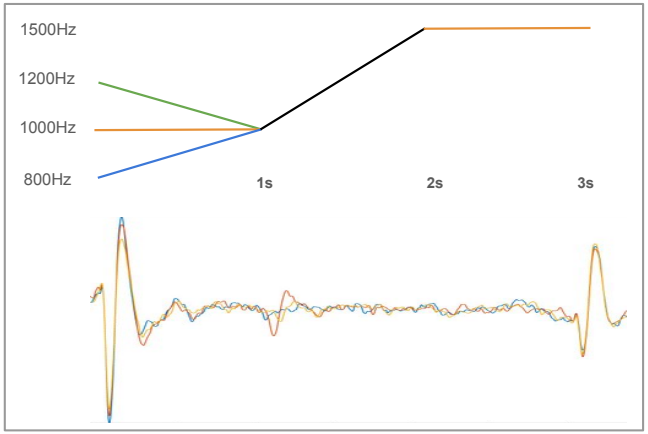


Models

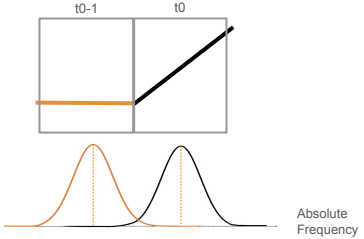
Model 1



The brain estimates average frequency over a time period, by comparing mean frequency of the current sound (t0) and that the recent past (t0-1). The brain expects mean frequency at t0-1 to match the one at t0. If it doesn't, this mismatch triggers a vertex in the neural response.



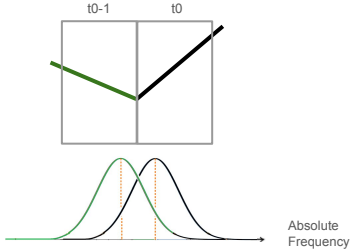
Case 1



mean flat f ≠ mean test ramp f:

There is a difference between the mean frequency at t0 and the mean frequency of the preceding bin (t0-1) → Little overlap of the frequency distributions → large difference in expected vs. actual input → brain sees this as "unexpected" → vertex elicited (**as seen in our observations**)

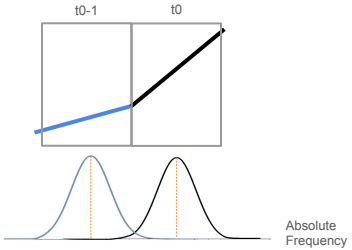
Case 2



mean descending f = mean test ramp f:

There is little difference between the mean frequency at t0 and the mean frequency of the preceding bin (t0-1) → Partial overlap of the frequency distributions → Minimal difference in expected vs. actual input → brain sees this as "not unexpected" → vertex not elicited (**as seen in our observations**)

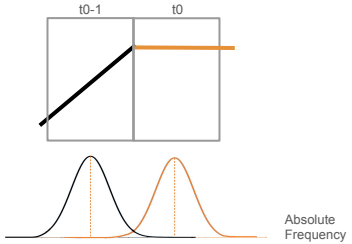
Case 3



mean rising f ≠ mean test ramp f:

Suggests we should see a vertex elicited at 1s. Does not explain why we **do not** see this in our observations.

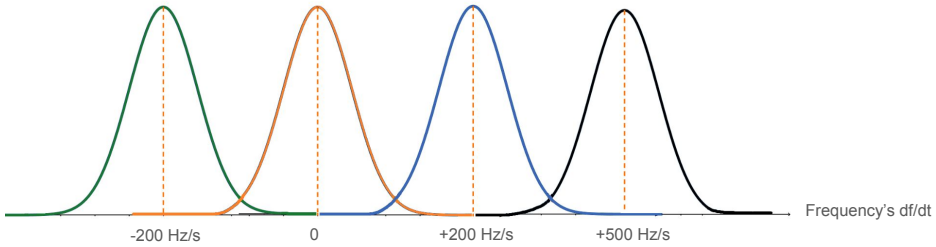
Case 4



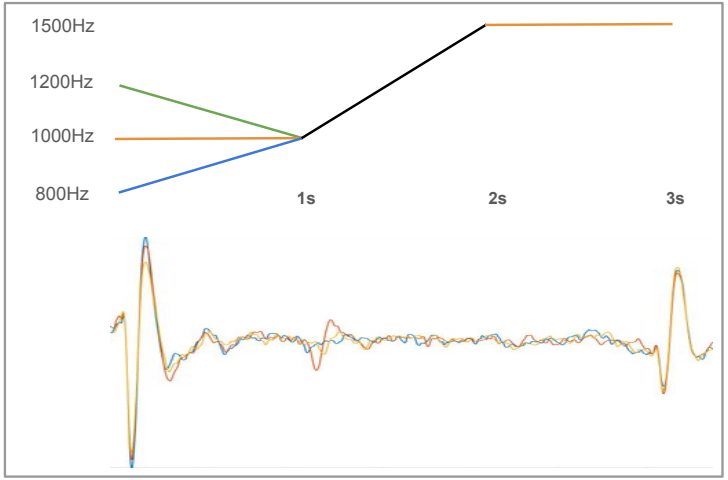
mean test ramp f ≠ mean plateau f:

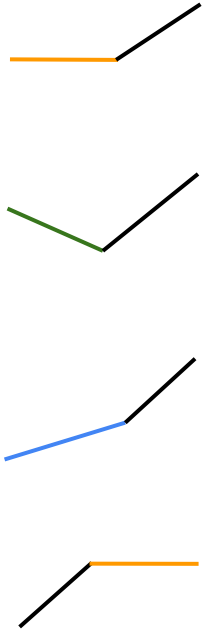
Suggests we should see a vertex elicited at 2s. Does not explain why we **no not** see this in our observations.

Model 2



Stimuli are encoded not as exact values, but distributions: the brain doesn't treat a slope of +200 Hz/s as a single point. Instead, it has some uncertainty, seen here as a Gaussian. It's a probabilistic comparison rather than a binary one.





In general

- If the df/dt distribution of the test ramp overlaps with that of the reference → the brain sees the test ramp as "expected" → no VP
- If the test ramp df/dt is far from the reference (distributions don't overlap) → prediction error → VP is elicited

Case 1: **Flat** (0 Hz/s) → **Test** ramp (+500 Hz/s)

- Distributions: minimal overlap
- Large df/dt shift (from 0 to fast rising) → Prediction error → vertex observed → **matches empirical data**

Case 2: **Descending** (–200 Hz/s) → **Test** ramp (+500 Hz/s)

- Distributions: minimal overlap
- Large df/dt shift (from falling to rising) → Strong prediction error → vertex observed → **does not match empirical data**
- Note: Could match empirical data if signs (-/+) were not significant in this case

Case 3: **Rising** (+200 Hz/s) → **Test** ramp (+500 Hz/s)

- Distributions: some overlap
- Both are rising, but test has steeper slope
- The magnitude of slope increases, but maybe not enough to register as a surprising change → No vertex observed → **matches empirical data**

Case 4: **Test** ramp (+500 Hz/s) → **Plateau** (0 Hz/s)

- Distributions: minimal overlap
- Moderate df/dt shift (rising to flat) → Prediction error → vertex is elicited → **does not match empirical data**
- Note: Could match empirical data if we suppose that this model has a directionality (only goes in the direction of negative to positive)